

AYMANE AIT DADS

ML Intern - Spacecraft Telemetry | PyTorch · Scikit-learn · Anomaly Detection

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Hiring Team - Internship - Leveraging Machine Learning for Spacecraft Telemetry

Innovative Solutions In Space · Delft, Netherlands

Dear ISISPACE Hiring Team,

Your internship calls for someone who can **develop and test machine learning models for anomaly detection** on real-world engineering data - at Orange Maroc, I built exactly that: an **end-to-end ML pipeline on 100K+ network telemetry records** combining Random Forest and KMeans, reducing manual analysis time by roughly 60% and directly informing commercial decisions.

On the **feature engineering pipeline** side, I applied cross-validated predictive maintenance modeling on live telemetry, then delivered executive-ready **Power BI dashboards** to senior management - covering the full arc from raw ingestion to communicated insight. Separately, I built a **transformer autoencoder** for unsupervised anomaly detection on industrial time-series audio (AUC 0.80, no labeled anomalies required), benchmarking Mel spectrogram representations under a DCASE-style evaluation protocol - a setup directly analogous to flagging faults in satellite subsystem streams.

ISISPACE operates its own Mission Control Centre and holds multi-mission telemetry datasets that most ML practitioners never get near. The chance to build a proof-of-concept anomaly detection tool on that data - where model outputs affect real satellite operations - is precisely the kind of production-grounded problem I want to work on. I would welcome the opportunity to discuss how my background fits this project.

Sincerely,

AYMANE AIT DADS